

Team Norms: **Promoters/Barriers**

Group: _____ Team _____ Session L-3-1 Date: _____

0. **Who is absent today:** member #:

A. Results of Silent Brainstorming (how many ideas did each member contribute?)

Member	# 1	# 2	# 3	# 4	# 5	Total	Ideas Remaining After Boggle
Promoters							
Barriers							

B-1. # of categories of PROMOTERS _____, maximum _____, minimum _____

B-2. # of categories of BARRIERS _____, maximum _____, minimum _____

C. Modified Multi-Voting: *STEP ONE: Each person uses one symbol for each vote he makes*

Member	# 1	# 2	# 3	# 4	# 5
symbol					

D. List top ideas in separate sheets:

	PROMOTERS	BARRIERS
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Summary of Team DNB Assessments

KEEP IN DNB

Must be in hardcopy

2026

Team _____

At the conclusion of each Peer Assessment activity, the team has the responsibility of entering the results of the Peer Assessment in the following table. The results of each assessment (found on the last page of the checklist) are entered in the appropriate columns for the checklist row. Grayed boxes will not have any entries. **The Laboratory Faculty will fill in the Final Assessment columns and reserve the right to change a Peer Assessment, either up or down.**

Checklist	Design Notebook Assessments				
	Initial Submission			For instructor use only	
	self	peer	inst	After the Re-Submit	Conclude: the Final Assessment
Zeroth					
First					
Second					
Third					
Fourth					
Fifth					
Final Submission					

Resubmission of Design Notebook Work Checklist starting 481

(case of NI/NCE/NS in the first submission)

Team 's Work Work Product _____ Submission Date / /2026

Reassessed by Team _____ Instructor _____

How to Resubmit Work for Reassessment: Instructions for the team who OWNS the work

If the notebook work has been assessed as NI, NCE, or NS the team must use the following process when resubmitting the work for reassessment. Failure to follow the steps of this process will result in the issuance of a Self-Regulation Lapse to the team, even if the assessed work meets expectations.

1. Review the checklist from the original assessment to see what areas need improvement. You must improve **all the No's** and at least **half of the Weak's** to have the work reassessed as Meets.
2. Prepare the improvements to your work and have them ready for the assigned time for reassessment by the peer-team prior to the assigned class for resubmission (see the course calendar for the notebook assessment dates).
3. On the original assignment checklist indicate, in the **Yes** or **Wow** boxes next to the topics improved, the **page numbers** in the notebook where you made the improvements (some checklist may have special columns for this).
4. Place this checklist (which is completed by you and will be completed by the assessor) **ON THE TOP** of the original checklist.

How to assess this work: Instructions for the peer-team (originally assessed the work)

5. Review each item with no or weak sign as assessed by the instructor
6. Review each correction as stated in items (3).
7. Place the summary of your assessment in the table below:

there were	 No(s)	and Weaks
Now they are	 No(s)	and Weaks

8. Place your final assessment in the left two columns of the following table (the yes & NO cells)

Yes	No	Self-Regulation Issues	Lapse
		1. Was the material resubmitted when due? (if NO a lapse will be issued by the instructor)	
		2. Was the work assembled in the correct order? (if NO a lapse will be issued by the instructor)	
		3. Were there page numbers on the original checklist showing where the improved work is located? (if NO a lapse will be issued by the instructor)	
Total lapses			

Results of Re-Assessment (to be issued by the instructor)

Work initially graded as NI are not subjected to NCE in the resubmission

M	If all the No's and at least half of the Weak's are corrected
NI	If some No's and/or less than half of the Weak's are corrected AND the original case was NI
NCE	If some No's and/or less than half of the Weak's are corrected AND the original case was NCE
Instructor initial:	
Total lapses	

The CORRECTION version Checklist for Assignment L-2

Team _____ Work Products Being Evaluated:

*The team must answer/comment all cells marked by (**)*

Work Evaluated by (self) _____ Assessment Symbol/Color _____ Date _____

Work Evaluated by (inst) _____ Assessment Symbol/Color _____ Date _____

Yes	No	Self-Regulation Issues: (Full lapse will be issued (to each member) for each NO)
		A. The assignment was uploaded to the Blackboard in MS-word-format prior to the due date
		B. The assignment was not uploaded to the Blackboard in MS-word-format prior to the end of the one week permitted for late submission
		C. Only one resubmission of the assignment is on the BB (neither zero nor multiple)
		D. The assignment is in four pages (not papers) or less
		E. The two attached checklists were <u>completely filled out</u> and stapled to the top of the rest of the assignment-work in order and from the left top corner
		F. There are no other Self-Regulation problems (if No state problem)
		Enter a 1 in the box to the left if there are any No's
Comments on any of the No's		

NA	Yes	No	Expected Features (mark yes or no for the points that need to be corrected; otherwise mark NA)
			1.The work satisfies the PTW checklist
			2.It is clear that the work starts with a set of ten promoters and a set of ten barriers from each team member
			3.There are evidence(s) that the team used Boggle Method to eliminate the duplicated ideas ** (Name the evident(s) the team used)
			4.There are evidence(s) that the team met after class to formulate the norms (agenda-minutes ... etc.) ** (Name the evident(s) the team used)
			5.It is clear how many promoters were selected to formulate the norms ** (How many promoters are used/selected)
			6.There are operational statement(s) for each selected norm (an action, response, or behavior) that will be used to achieve the desired result.
			7.It is clear how many barriers were selected to formulate the norms

The CORRECTION version Checklist for Assignment L-2

			** (state how many barriers are used)
			8. There is at least one prevention action for each selected barrier.
			9. There is at least one contingency action for each selected barrier.
			Comments on any of the No's

NA	Ok	Weak	Revealed Features (mark OK or Weak for the points that need to be corrected; otherwise mark NA)
			Rate the quality of the introduction/context of the entire assignment
			Rate the quality of the operational statements
			Rate the quality of the preventive actions
			Rate the quality of the contingency actions
			Rate the quality of the scenario used in presenting the entire assignment
			Rate the quality of the discussion of the entire assignment

Result summary

The self-regulation issues that marked No		Count the number of No in the expected features	WOWs	Weaks
Count	Equivalent Lapses			

FINAL ASSESSMENT

M	all <u>Yes</u> in the selected “expected features” (up to 1 Lapse is permitted here)			
NI	the sum of the number of NOs in the expected features and the number of Weaks is 3 or less.			
NCE	Is given in any other situation including			
	Total SR-Lapses			
The final conclusion rules: (1): If the original assignment was graded as NI; this correction cannot be granted less than NI. (2): If the original assignment was graded as NCE; this correction can be granted NI or NCE.				
The final grade is (mark one)	M	NI	NCE	+ lapses

***** close *****

Assignment # L4 (2026) (The Capstone Design Project)

PI (3.2)

Situation Description

High school students do not know enough information about manufacturing and design processes. This is a big disadvantage especially for engineering-students because they decided to pursue engineering as a career. They join engineering colleges where education is highly theoretical and emphasizes math and science. This emphasis is based primarily on the assumption that engineers are likely to learn the more applied portion of their field on the job while they are unlikely to learn math and science on the job.

The Project Learning Objectives

Many Learning Objectives can be accomplished and demonstrated as you work on this project. Not all of these objectives can be achieved in each of the project assignments but by the end of the project, your team should have achieved (and demonstrated achievement) for all of the following objectives:

1. Prepare High school students for engineering practice through the curriculum culminating in a **real-life** design experience based on the knowledge and skills acquired in earlier course work and incorporating engineering standards and realistic constraints that include most of the following considerations: economic; environmental; sustainability; manufacturability; ethical; health and safety; social; and political
2. Achieve some knowledge about how a real engineered product works
3. Achieve some awareness about how a real engineered product is manufactured
4. Demonstrate the ability to Define a Problem, Generate Solutions, make criteria-based Decisions, and Plan
5. Demonstrate ability to create a document and process
6. Demonstrate ability to function as a productive team
7. Demonstrate ability to manage a project
8. Demonstrate ability to create and maintain a Design Notebook
9. Demonstrate ability to assess the quality of work in a Design Notebook

Project Assignment Details

The Project is divided into five assignments as follows: -

- a. Problem Definition,
- b. Idea Generation,
- c. Artifact Selection,
- d. Implementation,
- e. Evaluation, Testing an Assembly Plan

For each of these assignments; you will find:

- **The Assignment** section where the specifics of the problem are given,
- **The Work In The Notebook** section where the expected organization of the work in the notebook is discussed, and
- **The Assessment** section where you are told which checklist will be used to assess the work.

It is recommended to read these pages carefully several times; till you understand the contents.

How Many Sandwiches?

As you start to work on your project you will begin to wonder just how many sandwiches are going to be required in presenting the work. The following three examples below show the sandwich structure applied more and more frequently. Your team will have to do at least what is shown in the second example. The third example shows what you might consider for assignment L4a; similar sub-sandwich structures may be reasonable for all of the assignments. The following two heuristics may help you decide when additional sub-sandwiches would be helpful:

1. if the work contains significantly different tasks consider sub-sandwiches for each task
2. if the work is only **two** to **five** pages long use only one sandwich for the work

I. The Big Picture

Project 1 Context

Assignment L4 Work

Project 1 Discussion

II. A little more Detail (minimum)

Project 1 Context

L4a *Context*

L4a Work

L4a *Discussion*

L4b *Context*

L4b Work

L4b *Discussion*

L4c *Context*

L4c Work

L4c *Discussion*

L4d *Context*

L4d Work

L4d *Discussion*

L4e *Context*

L4e Work

L4e *Discussion*

Project 1 Discussion

III. Yet Even More Detail

Project 1 Context

L4a *Context*

L4a Task1 *Context*

Task 1 Work

L4a Task 1 *Discussion*

L4a Tasks 2 & 3 *Context*

Tasks 2 & 3 Work

L4a Tasks 2 & 3 *Discussion*

L4a *Discussion*

L4b *Context*

L4b Work

L4b *Discussion*

L4c *Context*

L4c Work

L4c *Discussion*

L4d *Context*

L4d Work

L4d *Discussion*

L4e *Context*

L4e Work

L4e *Discussion*

Project 1 Discussion

It is recommended to read these pages carefully several times; till you understand the contents.

Assignment "Team Norms" Checklist

Team _____ Work Products Being Evaluated: **Team Norms (the original assignment)**

Work Evaluated by (self ?) _____ Assessment Symbol/Color _____ Date _____

Work Evaluated by (inst.) _____ Assessment Symbol/Color _____ Date _____

The top **two lines**, the **self-regulation** issues part and the **expected features** part are to be completed and self-assessed prior to submitting the work.

Yes	No	Self-Regulation Issues: <i>(half lapse will be issued (to each member) for each NO)</i>
		A. The assignment was uploaded to the Blackboard in MS-word-format prior to the due date (if this item is marked NO; the final grade will be NS)
		B. There is one and only one Only one submission of the assignment on the Blackboard
		C. The assignment is in four pages (not papers) or less (<i>this does not include the appendices</i>)
		D. The two attached checklists are stapled to the top of the rest of the assignment-work from the left top corner and are in order.
		E. The top two lines, the tables of the self-regulation issues and the expected features (in this checklist) are self-assessed and completed prior to submission. This includes answering questions indicated by (**). No other part in this checklist is marked or assessed
		F. The PTW is completely filled out
		G. There are no other Self-Regulation problems (if No state the point here)
		H. Enter a 1 in the box to the left if there are any No's

Comments

Yes	No	Expected Features
		1. The work satisfies the PTW checklist
		2. It is clear that the work starts with a set of ten promoters and a set of ten barriers from each team member
		3. There are evidence(s) that the team used Boggle Method to eliminate duplicate in ideas ** (Name the evident(s) the team used)
		4. There are evidence(s) that the team met after class to formulate the norms (agenda-minutes ... etc.) ** (Name the evident(s) the team used)
		5. It is clear how many promoters were selected to formulate the norms ** (How many promoters are used/selected)
		6. There are operational statement(s) for each selected norm (an action, response, or behavior) that will be used to achieve the desired result.

Assignment "Team Norms" Checklist

Yes	No	Expected Features
		7. It is clear how many barriers were selected to formulate the norms ** (state how many barriers are used)
		8. There is at least one prevention action for each selected barrier.
		9. There is at least one contingency action for each selected barrier.
Comments on any of the No's		

Wow	Ok	Weak	Revealed Features (for instructor use only)
			Rate the quality of the introduction/context of the entire assignment
			Rate the quality of the operational statements
			Rate the quality of the preventive actions
			Rate the quality of the contingency actions
			Rate the quality of the scenario used in presenting the entire assignment
			Rate the quality of the discussion of the entire assignment

Result summary (for instructor use only)

The self-regulation issues that marked No		Count the number of No in the expected features	WOWs	Weaks
Count	Equivalent Lapses			

Assessment (for instructor use only - In general; No piece of lapse is permitted with E)

E	Two WOWs and ZERO NO (the total sum of NO in this checklist is zero)
M	All Yes in the expected features with no single weak (up to 1 Lapse is permitted here)
NI	The sum of (the number of NO s in the expected features and the number of Weak) is 3 or less.
NCE	Is given in any other situation (except what is considered for NS)
NS	If item A is NO (in this case the assessment of the entire checklist is ignored)
	Total SR-Lapses

If the work was assessed as either **NI** or **NCE** the team **MAY** resubmit it (if applicable) on the time set by the course with appropriate checklist.

For instructor use only: the percentage of the plagiarism is acceptable	yes	No
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***** close *****